

QUANTITATIVE FINANCE

Case I - Duality Theory and Mean-Variance Analysis*

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For example, the famous **minimax theorem** of von Neumann (1928)¹, which was the starting point of game theory, is a kind of duality theorem.

Let $L : C_1 \times C_2 \rightarrow \mathbb{R}$ be a continuous function where $C_1 \subseteq \mathbb{R}^n$ and $C_2 \subseteq \mathbb{R}^m$ are compact convex sets. If L is a convex-concave function, that is, $\mathbf{x} \mapsto L(\mathbf{x}; \mathbf{y})$ is convex for fixed $\mathbf{y}$, and $\mathbf{y} \mapsto L(\mathbf{x}; \mathbf{y})$ is concave for fixed $\mathbf{x}$, then

This theorem provides a condition when the optimal values of the minimax and maximin problems are equal, and thus the order of optimizations can be flipped.

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A brief History

Shortly after the simplex method for linear programming was invented in 1947, the **existence of a duality theory** was hinted at by von Neumann.

In 1951, this duality theorem was extended to **convex programming** by Kuhn and Tucker (1951)², who also brought to light the connection between **duality and saddle points**.

²Kuhn, H. W. and A. W. Tucker, 1951. *Nonlinear programming*, Proceedings of the Second Berkeley Symposium on Mathematical Statistics and Probability, 1950 (Berkeley and Los Angeles), University of California Press, 481-492.

The Plan

- 1 We treat duality through the Lagrangian function and the minimax theorems for such functions.
- 2 Specifically, given a minimization problem, which we call the primal problem (P) , we devise a Lagrangian function $L(\mathbf{x}, \lambda)$, such that (P) is equivalent to the minimax problem $\inf_{\mathbf{x}} \sup_{\lambda} L(\mathbf{x}, \lambda)$. We then show that the minimax theorem holds:

$$\inf_{\mathbf{x}} \sup_{\lambda} L(\mathbf{x}, \lambda) = \sup_{\lambda} \inf_{\mathbf{x}} L(\mathbf{x}, \lambda). \quad (2)$$

We denote the RHS maximin problem $\sup_{\lambda} \inf_{\mathbf{x}} L(\mathbf{x}, \lambda)$ by (D) , and call it the dual of (P) .

- 3 The **strong duality theorem**, which holds for the pair (P) and (D) , yields the sweet fruits, e.g., the Lagrangian first-order condition for gradients.
- 4 We will see that **switching the order** of the inf and sup operations is a key simplification in model solving.
- 5 The replacement of min and max by inf and sup, respectively, are necessary as the objective function may be **unbounded** and hence the optimizer **unattainable** in general.

Setup

Consider a function $L : \mathcal{A} \times \mathcal{B} \rightarrow \mathbb{R}$, where $\mathcal{A}$ and $\mathcal{B}$ are arbitrary sets and L is an arbitrary function. We associate with L the primal and dual problems:

$$\inf_{\mathbf{x} \in \mathcal{A}} \sup_{\lambda \in \mathcal{B}} L(\mathbf{x}, \lambda), \quad (P)$$

$$\sup_{\lambda \in \mathcal{B}} \inf_{\mathbf{x} \in \mathcal{A}} L(\mathbf{x}, \lambda). \quad (D)$$

We denote the optimal values of (P) and (D) by $\inf(P)$ and $\sup(D)$, respectively.

Weak Duality Theorem

If (P) and (D) are the associated primal and dual programs with L , then

$$\sup(D) \leq \inf(P). \quad (3)$$

Proof.

For all $\mathbf{y} \in \mathcal{A}$ and $\lambda \in \mathcal{B}$,

$$\inf_{\mathbf{x} \in \mathcal{A}} L(\mathbf{x}, \lambda) \leq L(\mathbf{y}, \lambda). \quad (4)$$

This implies that

$$\sup_{\lambda \in \mathcal{B}} \inf_{\mathbf{x} \in \mathcal{A}} L(\mathbf{x}, \lambda) \leq \sup_{\lambda \in \mathcal{B}} L(\mathbf{y}, \lambda), \text{ for all } \mathbf{y} \in \mathcal{A}. \quad (5)$$

Then, in turn,

$$\sup(D) = \sup_{\lambda \in \mathcal{B}} \inf_{\mathbf{x} \in \mathcal{A}} L(\mathbf{x}, \lambda) \leq \inf_{\mathbf{y} \in \mathcal{A}} \sup_{\lambda \in \mathcal{B}} L(\mathbf{y}, \lambda) = \inf_{\mathbf{x} \in \mathcal{A}} \sup_{\lambda \in \mathcal{B}} L(\mathbf{x}, \lambda) = \inf(P). \quad (6)$$



Discussions

The weak duality theorem already gives us some useful information concerning the problems (P) and (D) .

Feasibility of Optimization Problems

Let L be a Lagrangian function, and let (P) and (D) be the associated primal and dual programs with L . If the primal minimization problem (P) is **unbounded**, then the dual maximization problem (D) is **infeasible**.

Proof.

(P) is unbounded if and only if $\inf(P) = -\infty$. Then by the weak duality theorem, $\sup(D) = -\infty$ as well, which means that (D) has an empty feasible region. $\square$

Discussions

The difference in the objective values,

$$\inf(P) - \sup(D), \quad (7)$$

is called the *duality gap*. Recall that $\sup(D)$ is an underestimator of $\inf(P)$. Hence the gap will always be nonnegative.

It is of great importance to know when the duality gap is zero, that is, when the optimal objective values of the primal and dual problems coincide. An interesting case is given by the following Saddle Point Theorem.

A point $(\mathbf{x}^*, \lambda^*) \in \mathcal{A} \times \mathcal{B}$ is a saddle point of L if

$$L(\mathbf{x}^*, \lambda) \leq L(\mathbf{x}^*, \lambda^*) \leq L(\mathbf{x}, \lambda^*), \text{ for all } \mathbf{x} \in \mathcal{A} \text{ and } \lambda \in \mathcal{B}. \quad (8)$$

Saddle Point Theorem

Let $L : \mathcal{A} \times \mathcal{B} \rightarrow \mathbb{R}$ and $(\mathbf{x}^*, \lambda^*) \in \mathcal{A} \times \mathcal{B}$. The following conditions are equivalent:

- (a) $(\mathbf{x}^*, \lambda^*)$ is a saddle point of $L(\mathbf{x}, \lambda)$.
- (b) $\mathbf{x}^*$ is a solution of (P) , λ^* is a solution of (D) , and $\min(P) = \max(D)$, that is,

$$\min_{\mathbf{x} \in \mathcal{A}} \sup_{\lambda \in \mathcal{B}} L(\mathbf{x}, \lambda) = \max_{\lambda \in \mathcal{B}} \inf_{\mathbf{x} \in \mathcal{A}} L(\mathbf{x}, \lambda), \quad (9)$$

where the minimization on the LHS is achieved at $\mathbf{x}^*$, and the maximization on the RHS is achieved at λ^* .

Furthermore, if either (a) or (b) is satisfied, then the common optimal value of (P) and (D) must be $L(\mathbf{x}^*, \lambda^*)$.

Note that if the minimizer(maximizer) is attainable, then we can put $\min(\max)$ in place of $\inf(\sup)$.

Proof

The Saddle Point Theorem is very important, but its proof turns out to be pellucid.

Proof.

Suppose that (a) holds. According to the definition of a saddle point, we have

$$\sup_{\lambda \in \mathcal{B}} L(\mathbf{x}^*, \lambda) = \max_{\lambda \in \mathcal{B}} L(\mathbf{x}^*, \lambda) = L(\mathbf{x}^*, \lambda^*) = \min_{\mathbf{x} \in \mathcal{A}} L(\mathbf{x}, \lambda^*) = \inf_{\mathbf{x} \in \mathcal{A}} L(\mathbf{x}, \lambda^*).$$

Then,

$$\begin{aligned} \inf_{\mathbf{x} \in \mathcal{A}} \sup_{\lambda \in \mathcal{B}} L(\mathbf{x}, \lambda) &\leq \sup_{\lambda \in \mathcal{B}} L(\mathbf{x}^*, \lambda) = L(\mathbf{x}^*, \lambda^*) = \inf_{\mathbf{x} \in \mathcal{A}} L(\mathbf{x}, \lambda^*) \\ &\leq \sup_{\lambda \in \mathcal{B}} \inf_{\mathbf{x} \in \mathcal{A}} L(\mathbf{x}, \lambda) \leq \inf_{\mathbf{x} \in \mathcal{A}} \sup_{\lambda \in \mathcal{B}} L(\mathbf{x}, \lambda). \end{aligned}$$

Because the first and the last terms are exactly the same, we must have equalities throughout. This proves (b) and the fact that the common optimal value of (P) and (D) equals $L(\mathbf{x}^*, \lambda^*)$. □

Proof - Cont'd

Proof.

Conversely, suppose that (b) holds. Then (9) gives

$$\sup_{\lambda \in \mathcal{B}} L(\mathbf{x}^*, \lambda) = \inf_{\mathbf{x} \in \mathcal{A}} L(\mathbf{x}, \lambda^*), \tag{10}$$

which in turn implies

$$L(\mathbf{x}^*, \lambda^*) \leq \sup_{\lambda \in \mathcal{B}} L(\mathbf{x}^*, \lambda) = \inf_{\mathbf{x} \in \mathcal{A}} L(\mathbf{x}, \lambda^*) \leq L(\mathbf{x}^*, \lambda^*). \tag{11}$$

Because the first and the last terms are exactly the same, we must have equalities throughout. This proves that the common optimal value $\min(P) = \max(D) = L(\mathbf{x}^*, \lambda^*)$. Furthermore, for all $\mathbf{x} \in \mathcal{A}$ and $\lambda \in \mathcal{B}$,

$$L(\mathbf{x}^*, \lambda) \leq \sup_{\lambda \in \mathcal{B}} L(\mathbf{x}^*, \lambda) = L(\mathbf{x}^*, \lambda^*) = \inf_{\mathbf{x} \in \mathcal{A}} L(\mathbf{x}, \lambda^*) \leq L(\mathbf{x}, \lambda^*). \tag{12}$$

This proves (a). □

Discussions

The theorem reveals the intimate connection between saddle points and a pair of primal-dual optimization problems. This connection was first brought to light by Kuhn and Tucker (1951), inspired by the linear programming duality and the game theory results of von Neumann (1928).

Next, we will apply the result to nonlinear programming duality which is closely linked to the Markowitz problem.

Nonlinear Programming Duality

Consider the nonlinear program:

$$\begin{aligned} \min_{\mathbf{x}} \quad & f(\mathbf{x}), \\ \text{s.t.} \quad & g_i(\mathbf{x}) = 0, \quad i = 1, 2, \dots, m, \\ & \mathbf{x} \in \mathcal{D} \subseteq \mathbb{R}^n, \end{aligned} \quad (P0)$$

with the associated Lagrangian function

$$L(\mathbf{x}, \boldsymbol{\lambda}) = f(\mathbf{x}) + \sum_{i=1}^m \lambda_i g_i(\mathbf{x}), \quad (13)$$

where $\mathbf{x} \in \mathcal{D}$, and $\boldsymbol{\lambda} = (\lambda_1, \lambda_2, \dots, \lambda_m)^\top \in \mathbb{R}^m$ is the multiplier.

Nonlinear Programming Duality

Note that

$$\sup_{\lambda_i \in \mathbb{R}} \lambda_i g_i(\mathbf{x}) = \begin{cases} 0, & \text{if } g_i(\mathbf{x}) = 0, \\ +\infty, & \text{otherwise.} \end{cases} \quad (14)$$

Thus,

$$\sup_{\lambda \in \mathbb{R}^m} L(\mathbf{x}, \lambda) = \begin{cases} f(\mathbf{x}), & \text{if } g_i(\mathbf{x}) = 0, \ i = 1, 2, \dots, m, \\ +\infty, & \text{otherwise.} \end{cases} \quad (15)$$

Consequently, we can write (P0) in the equivalent minimax form

$$\inf_{\mathbf{x} \in \mathcal{D}} \sup_{\lambda \in \mathbb{R}^m} L(\mathbf{x}, \lambda). \quad (P)$$

We formulate the dual program to (P) by switching the order of inf and sup above,

$$\sup_{\lambda \in \mathbb{R}^m} \inf_{\mathbf{x} \in \mathcal{D}} L(\mathbf{x}, \lambda). \quad (D)$$

We call (P) a *convex program* or a *convex programming problem* if the objective function f is convex, the equality constraint functions $\{g\}_1^m$ are affine, and $\mathcal{D} \subseteq \mathbb{R}^n$ is a nonempty convex set. We leave strong duality in general convex programming to future studies.

Key results regarding the Lagrangian function in (13), which suffice for the mean-variance analysis here and utility maximization in Part III, are summarized in the following theorem.

Saddle Points of the Lagrangian

Lagrangian Duality

A point $(\mathbf{x}^*, \lambda^*)$ is a saddle point of the Lagrangian L , that is,

$$L(\mathbf{x}^*, \lambda) \leq L(\mathbf{x}^*, \lambda^*) \leq L(\mathbf{x}, \lambda^*), \text{ for all } \mathbf{x} \in \mathcal{D} \text{ and } \lambda \in \mathbb{R}^m, \quad (16)$$

if and only if

- (i) $\mathbf{x}^*$ is a (global) minimizer of (P) ,
- (ii) λ^* is a (global) maximizer of (D) ,
- (iii) $\min(P) = \max(D)$.

Furthermore, if $(\mathbf{x}^*, \lambda^*)$ is a saddle point, then

- (iv) $\min(P) = L(\mathbf{x}^*, \lambda^*) = \max(D)$.

This theorem is the direct result of the previous Saddle Point Theorem.

Since $(P0)$ and (P) are equivalent, we have

$$f(\mathbf{x}^*) = \min(P0) = \min(P) = L(\mathbf{x}^*, \lambda^*),$$

where the last equality holds under **zero duality gap**. Put differently, $\mathbf{x}^*$ solves $(P0)$.

An Important Corollary

The Gradient Condition

If $(\mathbf{x}^*, \lambda^*)$ is a saddle point, then

$$L(\mathbf{x}^*, \lambda^*) \leq L(\mathbf{x}, \lambda^*), \text{ for all } \mathbf{x} \in \mathcal{D}.$$

That is, $\mathbf{x}^*$ is a (global) minimizer of $L(\mathbf{x}, \lambda^*)$.

Now if the functions f and $\{g_i\}_1^m$ are differentiable and $\mathcal{D} \subseteq \mathbb{R}^n$ is open, the first-order necessary condition for a minimizer indicates

$$\mathbf{0} = \nabla_{\mathbf{x}} L(\mathbf{x}^*, \lambda^*) = \nabla f(\mathbf{x}^*) + \sum_{i=1}^m \lambda_i^* \nabla g_i(\mathbf{x}^*).$$

The “global” in the above is innocuous, because for a “local” minimizer, we can always redefine $\mathcal{D}$ as the neighbourhood in which the local minimizer obtains, and the results hold verbatim.

Discussions

Let's take a careful look at the necessary condition for the gradients. It is derived from the following inequality:

$$L(\mathbf{x}^*, \boldsymbol{\lambda}^*) \leq L(\mathbf{x}, \boldsymbol{\lambda}^*), \text{ for all } \mathbf{x} \in \mathcal{D}.$$

Note that [this inequality actually comes from the dual problem \(D\)](#). Specifically, suppose the maximization of (D) is achieved at $\boldsymbol{\lambda}^*$. Then, we have

$$\sup_{\boldsymbol{\lambda} \in \mathbb{R}^m} \inf_{\mathbf{x} \in \mathcal{D}} L(\mathbf{x}, \boldsymbol{\lambda}) = \inf_{\mathbf{x} \in \mathcal{D}} L(\mathbf{x}, \boldsymbol{\lambda}^*) \leq L(\mathbf{x}, \boldsymbol{\lambda}^*), \text{ for all } \mathbf{x} \in \mathcal{D}. \quad (17)$$

In sum, we can fix $\boldsymbol{\lambda}^*$ to examine the (marginal) behavior of $\mathbf{x}$, a privilege extended by duality, or the flipping of inf and sup. The primal problem does not enjoy such an intriguing characterization. See the next slide.

Discussions

If we take a similar look at the primal problem (P) and suppose the minimization of (P) is achieved at $\mathbf{x}^*$, then we have

$$\inf_{\mathbf{x} \in \mathcal{D}} \sup_{\boldsymbol{\lambda} \in \mathbb{R}^m} L(\mathbf{x}, \boldsymbol{\lambda}) = \sup_{\boldsymbol{\lambda} \in \mathbb{R}^m} L(\mathbf{x}^*, \boldsymbol{\lambda}) \geq L(\mathbf{x}^*, \boldsymbol{\lambda}), \text{ for all } \boldsymbol{\lambda} \in \mathbb{R}^m. \quad (18)$$

Together with differentiability, we can conclude that

$$\mathbf{0} = \nabla_{\boldsymbol{\lambda}} L(\mathbf{x}^*, \boldsymbol{\lambda}^*) = (g_1(\mathbf{x}^*), g_2(\mathbf{x}^*), \dots, g_m(\mathbf{x}^*))^\top.$$

However, we cannot get similar conditions for $\mathbf{x}$. Even if we first try solving the inner maximization, then

$$\inf_{\mathbf{x} \in \mathcal{D}} \sup_{\boldsymbol{\lambda} \in \mathbb{R}^m} L(\mathbf{x}, \boldsymbol{\lambda}) = \inf_{\mathbf{x} \in \mathcal{D}} L(\mathbf{x}, \boldsymbol{\lambda}^*(\mathbf{x})), \quad (19)$$

which is hard to proceed since $L(\mathbf{x}, \boldsymbol{\lambda}^*(\mathbf{x}))$ can be unbounded and thus generally indifferentiable.

Convexity and Sufficiency under Duality

Suppose f is $\mathcal{C}^2$ and convex on an open and convex set $\mathcal{D}$. Now we check sufficiency by showing that if a point $(\mathbf{x}^*, \boldsymbol{\lambda}^*)$ satisfies $\nabla_{\mathbf{x}} L(\mathbf{x}^*, \boldsymbol{\lambda}^*) = \nabla f(\mathbf{x}^*) + \sum_{j=1}^m \lambda_j^* \nabla g_j(\mathbf{x}^*) = \mathbf{0}$ and $g_i(\mathbf{x}^*) = 0, i = 1, 2, \dots, m$, then $\mathbf{x}^*$ is a global minimizer to the original minimization.

According to Lagrangian Duality, we only need to show that $(\mathbf{x}^*, \boldsymbol{\lambda}^*)$ is a saddle point of the Lagrangian L , that is,

$$L(\mathbf{x}^*, \boldsymbol{\lambda}) \leq L(\mathbf{x}^*, \boldsymbol{\lambda}^*) \leq L(\mathbf{x}, \boldsymbol{\lambda}^*), \text{ for all } \mathbf{x} \in \mathcal{D} \text{ and } \boldsymbol{\lambda} \in \mathbb{R}^m,$$

where

$$L(\mathbf{x}, \boldsymbol{\lambda}) = f(\mathbf{x}) + \sum_{i=1}^m \lambda_i g_i(\mathbf{x}).$$

For the first inequality, since $g_i(\mathbf{x}^*) = 0, i = 1, 2, \dots, m$, we have

$$L(\mathbf{x}^*, \boldsymbol{\lambda}^*) = L(\mathbf{x}^*, \boldsymbol{\lambda}) = f(\mathbf{x}^*), \quad \forall \boldsymbol{\lambda} \in \mathbb{R}^m.$$

And for the second inequality, if the constraint functions $(g_i(\mathbf{x}), i = 1, 2, \dots, m)$ are affine, then $\nabla^2 g_i(\mathbf{x}) = \mathbf{0}$ and we obtain

$$\nabla_{\mathbf{x}}^2 L(\mathbf{x}^*, \boldsymbol{\lambda}^*) = \nabla^2 f(\mathbf{x}^*).$$

The convexity of f ensures that the stationary point $\mathbf{x}^*$ minimizes $L(\mathbf{x}, \boldsymbol{\lambda}^*)$ on $\mathcal{D}$. That is

$$L(\mathbf{x}^*, \boldsymbol{\lambda}^*) \leq L(\mathbf{x}, \boldsymbol{\lambda}^*).$$

(What if f is $\mathcal{C}^1$?)

Sufficiency When f is C^1

Suppose f is C^1 and convex on an open and convex set $\mathcal{D}$. Now we check sufficiency by showing that $(\mathbf{x}^*, \lambda^*)$ is a saddle point of L .

The first inequality can be proved in the same way.

For the second inequality, we first note that if $f_1(\mathbf{x})$ and $f_2(\mathbf{x})$ are both convex on some set $\mathcal{X}$, then $h(\mathbf{x}) = f_1(\mathbf{x}) + f_2(\mathbf{x})$ is also convex on $\mathcal{X}$. (Check this result by yourself using the definition of convexity!)

If $g_i(\mathbf{x})$ is affine, it is convex, and so is $\lambda_i g_i(\mathbf{x})$, no matter the sign of λ_i . Then given λ , $L(\mathbf{x}, \lambda)$ is convex w.r.t. $\mathbf{x}$ on $\mathcal{D}$. Using the first order condition for convexity, for any point $\mathbf{x} \in \mathcal{D}$, we have

$$L(\mathbf{x}, \lambda^*) \geq L(\mathbf{x}^*, \lambda^*) + (\nabla_{\mathbf{x}} L(\mathbf{x}^*, \lambda^*))^\top (\mathbf{x} - \mathbf{x}^*) = L(\mathbf{x}^*, \lambda^*).$$

This proves the second inequality.

(What if the constraint functions $(g_i(\mathbf{x}), i = 1, 2, \dots, m)$ are not affine?)

Sufficiency under Non-Affine Constraint Functions

The first inequality can be proved in the same way.

For the second inequality, clearly, we no longer have $\nabla_{\mathbf{x}}^2 L(\mathbf{x}^*, \lambda^*) = \nabla^2 f(\mathbf{x}^*)$.

If $\sum_{i=1}^m \lambda_i^* g_i(\mathbf{x})$ is convex on $\mathcal{D}$ (note that the signs of λ_i^* s now matter), we know that given λ^* , $L(\mathbf{x}, \lambda^*)$ is still convex w.r.t. $\mathbf{x}$ on $\mathcal{D}$. Assume f and $\mathbf{g}$ are $\mathbf{C}^1$. Again, using the first order condition for convexity, for any point $\mathbf{x} \in \mathcal{D}$, we have

$$L(\mathbf{x}, \lambda^*) \geq L(\mathbf{x}^*, \lambda^*) + (\nabla_{\mathbf{x}} L(\mathbf{x}^*, \lambda^*))^\top (\mathbf{x} - \mathbf{x}^*) = L(\mathbf{x}^*, \lambda^*).$$

This proves the second inequality.

Inequality Constraints

We may write the Markowitz problem in a more general form as follows.

$$\begin{aligned}
 \min_{w_1, w_2, \dots, w_N} \quad & \frac{1}{2} \sum_{i=1}^N \sum_{j=1}^N w_i \sigma_{ij} w_j, \\
 \text{s.t.} \quad & \sum_{i=1}^N w_i \bar{r}_i \geq \bar{r}_p, \\
 & \sum_{i=1}^N w_i = 1.
 \end{aligned}$$

That is, we want to find the minimum variance portfolio that achieves an expected return of at least $\bar{r}_p$.

Duality with Inequality Constraints

Consider the nonlinear program:

$$\begin{aligned}
 \min_{\mathbf{x}} \quad & f(\mathbf{x}), & (P0) \\
 \text{s.t.} \quad & g_i(\mathbf{x}) = 0, \quad i = 1, 2, \dots, m, \\
 & h_j(\mathbf{x}) \leq 0, \quad j = 1, 2, \dots, l, \\
 & \mathbf{x} \in \mathcal{D} \subseteq \mathbb{R}^n,
 \end{aligned}$$

with the associated Lagrangian function

$$L(\mathbf{x}, \boldsymbol{\lambda}, \boldsymbol{\mu}) = f(\mathbf{x}) + \sum_{i=1}^m \lambda_i g_i(\mathbf{x}) + \sum_{j=1}^l \mu_j h_j(\mathbf{x}), \quad (20)$$

where $\mathbf{x} \in \mathcal{D}$, and $\boldsymbol{\lambda} = (\lambda_1, \lambda_2, \dots, \lambda_m)^\top \in \mathbb{R}^m$ and $\boldsymbol{\mu} = (\mu_1, \mu_2, \dots, \mu_l)^\top \in \mathbb{R}^l$ are the multipliers.

Duality with Inequality Constraints

Note that

$$\sup_{\lambda_i \in \mathbb{R}} \lambda_i g_i(\mathbf{x}) = \begin{cases} 0, & \text{if } g_i(\mathbf{x}) = 0, \\ +\infty, & \text{otherwise,} \end{cases} \quad (21)$$

and

$$\sup_{\mu_j \geq 0} \mu_j h_j(\mathbf{x}) = \begin{cases} 0, & \text{if } h_j(\mathbf{x}) \leq 0, \\ +\infty, & \text{otherwise.} \end{cases} \quad (22)$$

Duality with Inequality Constraints

Thus,

$$\sup_{\lambda \in \mathbb{R}^m, \mu \in \mathbb{R}_+^l} L(\mathbf{x}, \lambda, \mu) = \begin{cases} f(\mathbf{x}), & \text{if } g_i(\mathbf{x}) = 0, \ i = 1, 2, \dots, m, \\ & h_j(\mathbf{x}) \leq 0, \ j = 1, 2, \dots, l, \\ +\infty, & \text{otherwise.} \end{cases} \quad (23)$$

Consequently, we can write (P0) in the equivalent minimax form

$$\inf_{\mathbf{x} \in \mathcal{D}} \sup_{\lambda \in \mathbb{R}^m, \mu \in \mathbb{R}_+^l} L(\mathbf{x}, \lambda, \mu). \quad (P)$$

We formulate the dual program to (P) by switching the order of inf and sup above,

$$\sup_{\lambda \in \mathbb{R}^m, \mu \in \mathbb{R}_+^l} \inf_{\mathbf{x} \in \mathcal{D}} L(\mathbf{x}, \lambda, \mu). \quad (D)$$

We will not go into detail about this model, but point out two key results as natural extensions to our previous discussions.

The Gradient Condition

If $(\mathbf{x}^*, \lambda^*, \mu^*)$ is a saddle point, then

$$L(\mathbf{x}^*, \lambda^*, \mu^*) \leq L(\mathbf{x}, \lambda^*, \mu^*), \text{ for all } \mathbf{x} \in \mathcal{D}.$$

That is, $\mathbf{x}^*$ is a (global) minimizer of $L(\mathbf{x}, \lambda^*, \mu^*)$.

Now if the functions f , $\{g\}_1^m$, and $\{h\}_1^l$ are differentiable and $\mathcal{D} \subseteq \mathbb{R}^n$ is open, the first-order necessary condition for a minimizer indicates

$$\mathbf{0} = \nabla_{\mathbf{x}} L(\mathbf{x}^*, \lambda^*, \mu^*) = \nabla f(\mathbf{x}^*) + \sum_{i=1}^m \lambda_i^* \nabla g_i(\mathbf{x}^*) + \sum_{j=1}^l \mu_j^* \nabla h_j(\mathbf{x}^*).$$

The Complementarity Conditions

If $(\mathbf{x}^*, \lambda^*, \mu^*)$ is a saddle point, then the duality gap is zero, and

$$f(\mathbf{x}^*) = L(\mathbf{x}^*, \lambda^*, \mu^*) = f(\mathbf{x}^*) + \sum_{i=1}^m \lambda_i^* g_i(\mathbf{x}^*) + \sum_{j=1}^l \mu_j^* h_j(\mathbf{x}^*).$$

Since $g_i(\mathbf{x}^*) = 0$, we get

$$\sum_{j=1}^l \mu_j^* h_j(\mathbf{x}^*) = 0.$$

In turn, we get the well-known complementary conditions:

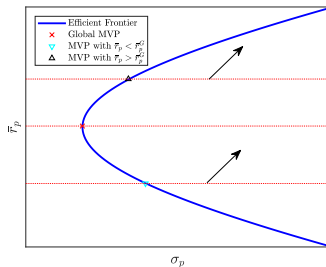
$$\mu_j^* \geq 0, \quad h_j(\mathbf{x}^*) \leq 0, \quad \mu_j^* h_j(\mathbf{x}^*) = 0, \quad j = 1, 2, \dots, l.$$

Reformulation of the Markowitz Problem

$$\begin{aligned}
 \min_{\mathbf{w} \in \mathbb{R}^n} \quad & \frac{1}{2} \mathbf{w}^\top \Sigma \mathbf{w}, \\
 \text{s.t.} \quad & \mathbf{w}^\top \mathbf{1}_n = 1, \\
 & \bar{r}_p - \mathbf{w}^\top \bar{\mathbf{r}} \leq 0,
 \end{aligned}$$

where $\mathbf{w}$ is the vector of portfolio weights, Σ is the variance-covariance matrix, $\mathbf{1}_n$ is a $n \times 1$ vector with all its components equal to 1, and $\bar{\mathbf{r}}$ is the vector of expected returns for individual assets.

Discussions on the Inequality Constraint



The inequality constraint is $\bar{r}_p - \mathbf{w}^\top \bar{\mathbf{r}} \leq 0$ or $\mathbf{w}^\top \bar{\mathbf{r}} \geq \bar{r}_p$.

The above figure gives the Minimum Variance Portfolios (MVPs) when $\mathbf{w}^\top \bar{\mathbf{r}} = \bar{r}_p$. (We already know how to solve for them.)

Now we relax the equality constraint to consider the inequality constraint $\mathbf{w}^\top \bar{\mathbf{r}} \geq \bar{r}_p$. Let $\bar{r}_p^G$ denote the expected return associated with the **Global MVP**.

- 1 $\bar{r}_p < \bar{r}_p^G$: The solution is clearly achieved when $\mathbf{w}^\top \bar{\mathbf{r}} = \bar{r}_p^G > \bar{r}_p$. The inequality is strict, and thus non-binding and can be removed from the minimization program.
- 2 $\bar{r}_p \geq \bar{r}_p^G$: The solution is achieved when $\mathbf{w}^\top \bar{\mathbf{r}} = \bar{r}_p$, as any higher return will be associated with a higher risk in this range. We can replace the inequality by the equality $\mathbf{w}^\top \bar{\mathbf{r}} = \bar{r}_p$.

The Complementary Conditions

Let $L(\mathbf{x}, \lambda, \mu) = f(\mathbf{x}) + \lambda(\mathbf{w}^\top \mathbf{1}_n - 1) + \mu(\bar{r}_p - \mathbf{w}^\top \bar{\mathbf{r}})$. The complementary condition associated with the inequality here is

$$\mu^* \geq 0, \bar{r}_p - \mathbf{w}^{*\top} \bar{\mathbf{r}} \leq 0, \mu^* (\bar{r}_p - \mathbf{w}^{*\top} \bar{\mathbf{r}}) = 0.$$

Then,

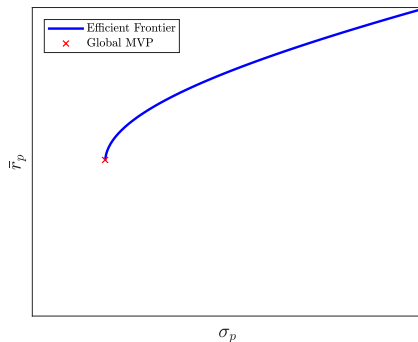
- 1 $\bar{r}_p < \bar{r}_p^G$: $\bar{r}_p - \mathbf{w}^{*\top} \bar{\mathbf{r}} < 0$ and thus $\mu^* = 0$. We have $L(\mathbf{x}^*, \lambda^*, \mu^*) = L(\mathbf{x}^*, \lambda^*)$.
The righthand side is the Lagrangian of the problem of finding the Global MVP:

$$\begin{aligned} \min_{\mathbf{w} \in \mathbb{R}^n} \quad & \frac{1}{2} \mathbf{w}^\top \Sigma \mathbf{w}, \\ \text{s.t.} \quad & \mathbf{w}^\top \mathbf{1}_n = 1. \end{aligned}$$

(If you try $\bar{r}_p - \mathbf{w}^{*\top} \bar{\mathbf{r}} = 0$, then we are back the case of two equality constraints. We know the solution **cannot** be better because the feasible set is now smaller.)

- 2 $\bar{r}_p > \bar{r}_p^G$: $\bar{r}_p - \mathbf{w}^{*\top} \bar{\mathbf{r}} = 0$. We go back to the familiar case of two equality constraints. (If you try $\bar{r}_p - \mathbf{w}^{*\top} \bar{\mathbf{r}} < 0$ and $\mu^* = 0$, you again get the Global MVP as the solution, which is now **infeasible** because $(\mathbf{w}^{*G})^\top \bar{\mathbf{r}} = \bar{r}_p^G < \bar{r}_p$.)
- 3 $\bar{r}_p = \bar{r}_p^G$: Either $\bar{r}_p - \mathbf{w}^{*\top} \bar{\mathbf{r}} = 0$ or $\mu^* = 0$ gives us the same solution of the Global MVP.

The New Efficient Frontier



Now the efficient frontier only contains the **upper half** above the global MVP. (The lower half is dominated by the global MVP which has a lower risk plus a higher return.)